$$F(6) = P(X \le 6) = \sum_{-\infty}^{6} P(X = x)$$

$$= P(X < 2) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6)$$

$$= 0 + \frac{1}{36} + \frac{4}{36} + \frac{10}{36} + \frac{12}{36} + \frac{9}{36} = 1.$$

Therefore the cumulative distribution function is

$$\begin{cases} 0 & \text{for} \quad -\infty < x < 2 \\ \frac{1}{36} & \text{for} \quad 2 \le x < 3 \\ \frac{5}{36} & \text{for} \quad 3 \le x < 4 \\ \frac{15}{36} & \text{for} \quad 4 \le x < 5 \\ \frac{27}{36} & \text{for} \quad 5 \le x < 6 \\ 1 & \text{for} \quad 6 \le x < \infty \end{cases}$$

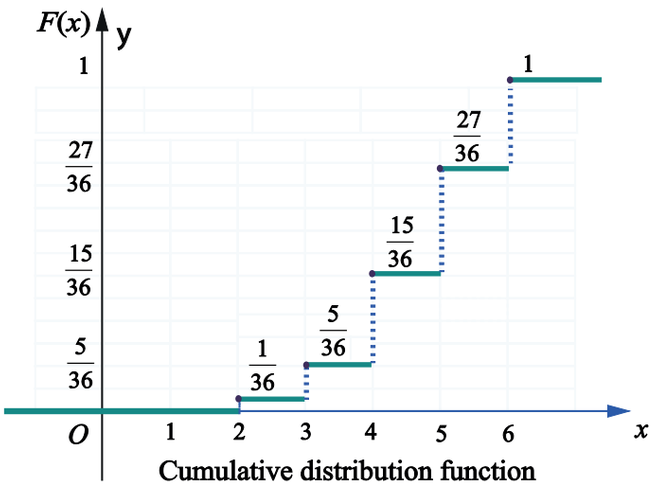


Fig. 11.9

(iii) $P(3 \le X < 6) = \sum_{x=3}^{5} P(X = x_i) = P(X = 3) + P(X = 4) + P(X = 5)$

$$= \frac{4}{36} + \frac{10}{36} + \frac{12}{36} = \frac{26}{36}.$$

(iv) $P(X \ge 4) = \sum_{x=4}^{\infty} P(X = x_i)$

$$= P(X = 4) + P(X = 5) + P(X = 6)$$

$$= \frac{10}{36} + \frac{12}{36} + \frac{9}{36} = \frac{31}{36}.$$

11.3.5 Probability Mass Function from Cumulative Distribution Function

For a discrete random variable X, the cumulative distribution function F has jumps at each of the x_i, and is constant between successive $x_i's$. The height of the jump at x_i is $f(x_i)$; in this way the probability at x_i can be retrieved from F.

Chapter 11 Probability Distributions-new.indd 191 29-05-2020 13:06:28